Room Name: Passive Reconnaissance

Objective: To understand passive reconnaissance and learn how to gather useful information about a target from publicly available sources without directly interacting with it.

Key Concepts Learned:

- Passive vs. Active Reconnaissance
- OSINT (Open-Source Intelligence)
- WHOIS and RDAP
- DNS Records and DNS Lookup
- Subdomain Discovery
- Certificate Transparency (CT) Logs
- Publicly Exposed Services
- Attack Surface Information Gathering

Tools and Websites Covered:

- **WHOIS:** Used to find domain registration details, registrar, registration dates, and name servers.
- **RDAP:** A modern alternative to WHOIS that provides domain registration information in structured JSON format.
- **nslookup:** Used to query DNS records such as A, MX, and TXT.
- **dig:** A DNS lookup tool that provides detailed information about DNS records.
- **DNSDumpster:** Used to discover subdomains, IP addresses, and related DNS information.
- **crt.sh:** Used to find subdomains through publicly available SSL/TLS certificate records.
- **Shodan:** Used to find publicly exposed devices, open ports, services, and server information.
- **Censys:** An alternative platform for exploring internet-exposed hosts and certificates.

Practical Work:

- Explored the difference between passive and active reconnaissance.
- Retrieved domain registration information using WHOIS and RDAP.
- Practiced DNS lookups using `nslookup` and `dig`.
- Explored different DNS record types, including A, AAAA, MX, CNAME, SOA, and TXT.
- Used DNSDumpster and crt.sh to identify publicly available subdomains.
- Explored Shodan to examine exposed hosts, services, open ports, and server banners.
- Learned how publicly available information can help identify an organization's attack surface.

Useful Commands:

```
# Retrieve domain registration details
whois example.com

# DNS lookup using nslookup
nslookup example.com

# Check specific DNS records
nslookup -type=MX example.com
nslookup -type=TXT example.com

# DNS lookup using dig
dig example.com A
dig example.com MX
dig example.com TXT

# Query DNS through a specific resolver
dig @1.1.1.1 example.com MX

# Retrieve domain information through RDAP
curl -s https://rdap.verisign.com/com/v1/domain/example.com | jq .
```

Important Observations:

- Passive reconnaissance uses public information without directly contacting the target.
- WHOIS and RDAP provide useful information about domain ownership and registration.
- DNS records help identify IP addresses, mail servers, and other domain-related information.
- DNSDumpster and crt.sh can reveal subdomains that may not appear in basic DNS lookups.
- Shodan helps identify publicly exposed services and their associated technologies.
- Information gathered during reconnaissance can help security professionals understand an organization's external attack surface.

Room Link: https://tryhackme.com/room/passiverecon

Status: Complete